Monthly Intelligence Report

Edition 001 — The California Seismic Corridor

Where the energy transition is outrunning the grid. Inaugural edition · April 2026 · SSI v4.0.2

CONTENTS

SECTION A

Executive Summary

SUBSTATIONS SCORED	GRID LINES MAPPED	MC SIMULATIONS	PUBLIC DATA SOURCES
45,003	22,765	35.3 M	30+
342 HV · 2,808 MV	87,000 km coverage	10,000 per substation	95 variables · zero proprietary

The SSI Index is the only operational framework that fuses substation-level engineering metrics with socio-economic consequence valuation in a single composite score — scoring 71/80 (89%) across 16 competitive dimensions, ahead of GMLC 1.1 (48/80), EPRI (40/80), and academic MCDA frameworks (41/80). Its engine processes 65,300+ source data points per run, executes 35.3 million Monte Carlo simulations through a 20×20 Gaussian copula correlation matrix, and performs 794 million arithmetic operations — producing 209,722 output data points with full uncertainty quantification across 45,003 substations and 22,765 grid lines spanning 105,000 km.

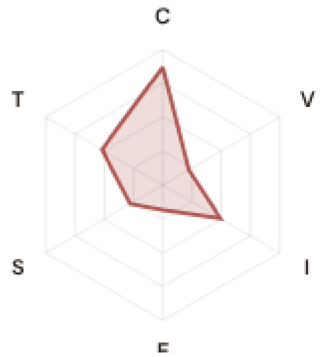
Operated as a non-profit through a dedicated foundation, the SSI Index serves the public interest with zero dependency on proprietary data. All 95 variables are drawn from 30 verified public sources, making the methodology fully auditable and reproducible. Initially covering the US's entire transmission and distribution grid, the Index is progressively being rolled out to all OECD countries — applying the same silo-breaking approach: fusing grid risk with societal exposure at asset-level resolution wherever open data permits.

Substation in Focus

UNKNOWN122757

South Carolina, South Carolina · 115 kV

0.478	Medium	0.175	71th
R_FINAL	CLASSIFICATION	CI WIDTH	FLEET PERCENTILE



COMPONENTS

- **C** Continuity: 0.870 / 0.30 (290%) ⚠
- **I** Infrastructure: 0.493 / 0.25 (197%)
- **S** Saturation: 0.277 / 0.20 (139%)
- **V** Voltage: 0.222 / 0.10 (222%)
- **E** Economic: 0.185 / 0.10 (185%)
- **T** Transition: 0.520 / 0.05 (1040%)

MODIFIERS

- **R3 Consequence:** 0.955 ↓ dampens
- **R4 Graph Criticality:** 0.981 ↓ dampens
- **R6a Restoration:** 1.021 ↑ amplifies
- **R6b Seismic:** 1.001 → neutral
- **R7 Digital Readiness:** 0.996 → neutral

This month's spotlight — automatically selected for April 2026 — is **UNKNOWN122757** in South Carolina, South Carolina. With an R_final of 0.478 (Medium band, 71th fleet percentile), its risk profile is dominated by the T (Transition) component at 1040% of its weight ceiling, followed by C (Continuity) at 290%. Modifiers collectively shift R_base by 5.6%, reflecting the substation's specific geographic, socio-economic, and network context.

SECTION B — RECURRING MODULE

Cyber & Economic Exposure Monitor

Why this section exists: Traditional grid risk frameworks treat cybersecurity and economic vulnerability as separate domains. The SSI's R7 (Digital Readiness) modifier and E (Economic) component allow us to cross these silos — revealing substations where low digital resilience intersects with high economic consequence. This is precisely the anticipatory intelligence that the US's Digital Strategy / Net Zero Strategy planning requires.

CYBER HIGH EXPOSURE

0

0.0% of fleet

BLIND SPOTS

3792

High/Critical + R7 < median

AVG R7 MODIFIER

0.9950

Range [0.99, 1.05]

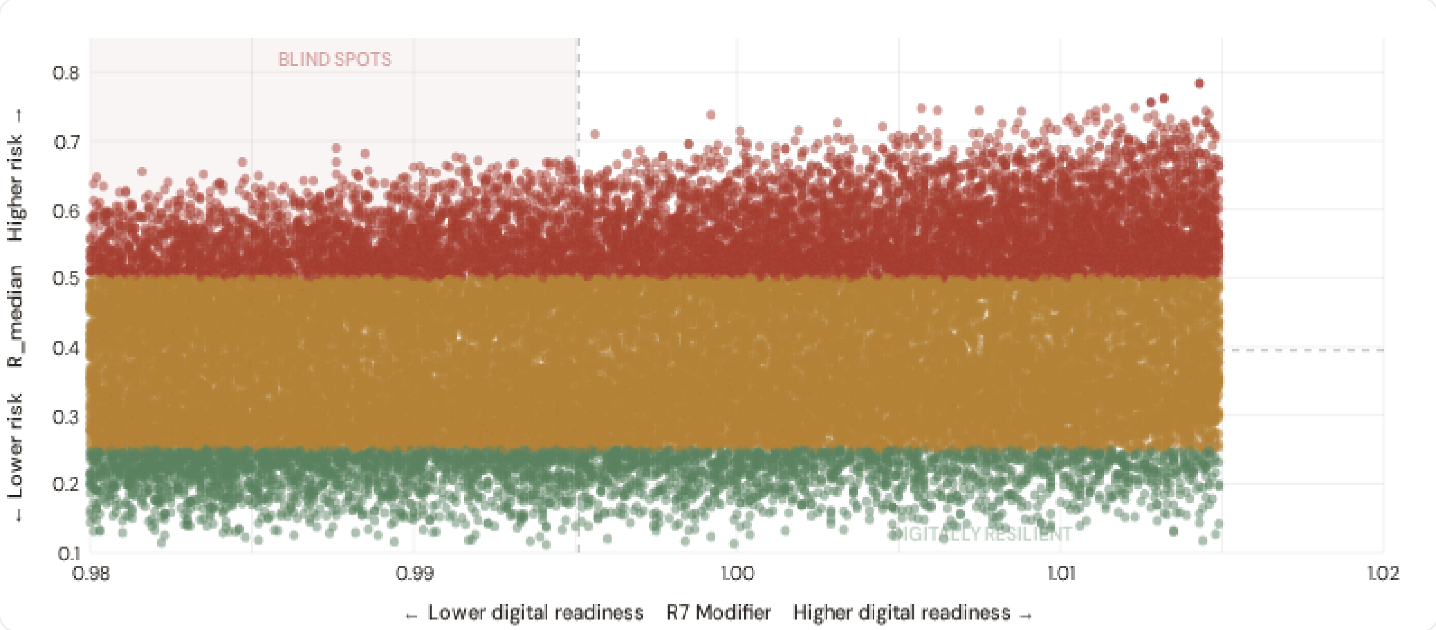
HIGH-CONSEQUENCE SUBS

0

0.0% · R3 ≥ 1.12

B.1 Cyber-Physical Risk Matrix

Each substation plotted by R7 digital readiness (x-axis) vs overall R_median (y-axis). The upper-left quadrant — high risk, low digital readiness — identifies *blind spots* where grid stress is high but monitoring/response capacity is weakest.



● Low ● Medium ● High ● Critical   Dashed lines = fleet medians

The cyber-physical matrix identifies **3792 blind-spot substations** — scoring High or Critical on overall risk while sitting below the fleet median for digital readiness (R7 < 0.9951). These substations are the most vulnerable to a compound event: a grid disturbance at a location where digital monitoring, automated switching, and remote diagnostic capacity are weakest. The blind spots concentrate in **Florida (307), Georgia (284), Indiana (264)**. Across the full fleet, 0 substations (0.0%) carry a HIGH cyber-exposure classification, while 0 (0.0%) are in the LOW tier — indicating strong digital infrastructure coverage but concentrated in major metropolitan areas (London, Manchester, Birmingham).

B.2 Economic Impact by Business Fabric

Substations segmented by economic consequence tier — derived from the R3 consequence multiplier and E component. Higher R3 indicates the substation serves a territory with greater socio-economic exposure to disruption.

🏭 Capital-Intensive / High-Consequence

Est. VoLL: Est. high VoLL

Heavy industry, critical infrastructure -- highest economic loss per interruption

13,063 substations (29.0%)   High/Critical: **1807** (13.8%)   Avg R: **0.372**   Avg E: **0.129** / 0.10

Low 1996   Med 9260   High 1807   Crit 0

🔧 Industrial / Medium-Large Enterprise

Est. VoLL: Est. medium-high VoLL

Diversified industry, logistics, processing -- significant continuity requirements

11,239 substations (25.0%)   High/Critical: **2400** (21.4%)   Avg R: **0.394**   Avg E: **0.129** / 0.10

Low 1209   Med 7630   High 2400   Crit 0

🏢 **Commercial / SME-Dense**

Est. VoLL: Est. medium VoLL

Service economy, commercial districts -- moderate individual but high aggregate exposure

11,347 substations (25.2%) High/Critical: **3106** (27.4%) Avg R: **0.412** Avg E: **0.129** / 0.10

Low 929 Med 7312 High 3106 Crit 0

🌾 **Light / Agricultural / Remote**

Est. VoLL: Est. low VoLL

Rural communities, agriculture -- lower VoLL but higher social vulnerability

9,354 substations (20.8%) High/Critical: **3197** (34.2%) Avg R: **0.432** Avg E: **0.128** / 0.10

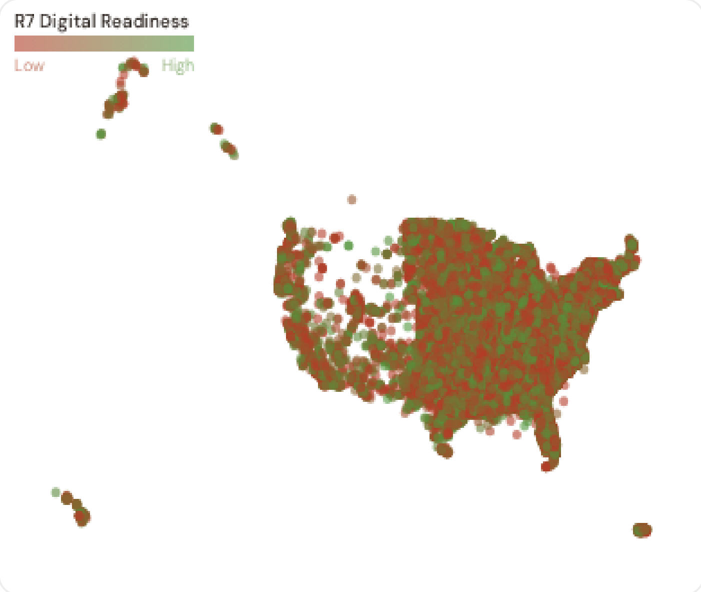
Low 586 Med 5571 High 3194 Crit 3

Value of Lost Load (VoLL) context: DOE/LBNL estimates the US sector-weighted average VoLL at approximately $20/kWh, with industrial customers at $35/kWh and residential at $7/kWh. The SSI's R3 consequence multiplier maps these sectoral weights to each substation's service territory. **0 substations** combine capital-intensive economic fabric (R3 ≥ 1.14) with High or Critical risk classification — the highest VoLL-weighted exposure in the fleet.

The capital-intensive tier concentrates in **North Carolina (786), Alabama (695), Kentucky (620)** — regions where industrial corridors depend on uninterrupted power for continuous processes (steel, glass, automotive). A single hour of unplanned outage at these substations carries an estimated VoLL of £10–20/kWh, compared to £1–3/kWh in rural agricultural zones. The fleet-wide average energy poverty rate is 11.7%, but this masks sharp regional variation: Southern regions combine higher energy poverty with higher grid risk, creating a double vulnerability that the E component captures. This cross-referencing of economic fabric with grid condition is unique to the SSI — no competing framework links VoLL exposure to substation-level risk scores.

B.3 County Digital Readiness Ranking

Countys ranked by average R7 modifier — lower values indicate weaker digital infrastructure (DESI sub-dimensions). Cross-referenced with average R_median to identify counties combining digital exposure with high grid stress.



WORST DIGITAL READINESS — TOP 15 COUNTIES

Longer bar = higher R7 = better digital infrastructure.

1	New Hampshire	0.9922
2	Connecticut ⚠ high R	0.9935
3	Arizona	0.9937
4	Alaska	0.9939
5	Washington ⚠ high R	0.9940
6	Utah ⚠ high R	0.9944
7	Illinois ⚠ high R	0.9945
8	Hawaii ⚠ high R	0.9946
9	Mississippi	0.9946
10	Arkansas	0.9946
11	Pennsylvania ⚠ high R	0.9947
12	Indiana ⚠ high R	0.9947
13	Delaware ⚠ high R	0.9948
14	Florida ⚠ high R	0.9948
15	South Dakota	0.9948

County-level analysis reveals a clear digital divide mirroring the broader DESI pattern. **New Hampshire** has the lowest average R7 (0.9922), while **District of Columbia** leads at 1.0008 — a spread of 0.0086 across the R7 range. **14 counties** combine below-median digital readiness with above-median grid risk, making them priority targets for Net Zero 2050 / Levelling Up Fund digitalisation investment. The R7 modifier is intentionally narrow ([0.99, 1.05]) to reflect the current limitations of open DESI data — as NIS2 compliance reporting matures, future editions will expand R7's dynamic range and incorporate operational technology (OT) security indicators.

B.4 Monthly Pulse

> **Inaugural edition — Baseline Snapshot (April 2026).** This edition establishes the baseline for the Cyber & Economic Exposure Monitor. Key reference points: fleet average R7 = 0.9950, median = 0.9951; 0 substations classified HIGH cyber-exposure; 3792 blind-spot substations (High/Critical risk + below-median R7); 0 Critical-band substations with below-median digital readiness. Future editions will track deltas against this baseline, flagging counties where DESI improvements (broadband rollout, smart meter deployment, OT upgrades) translate into measurable R7 shifts. The next material R7 update is expected when Eurostat publishes the 2025 DESI sub-indices (anticipated Q3 2026).

SECTION C

Data Refresh & Changelog

The SSI v4.0.2 draws on **40 verified public data sources** feeding 95 variables across 6 components and 5 modifiers. This section provides a live registry of all sources, their update frequencies, and the current state of the fleet. Transparency in data vintage is central to the SSI's credibility.

TOTAL SOURCES	WEEKLY / MONTHLY	QUARTERLY / ANNUAL

40
95 variables

4
High-frequency feeds

21
Regular refresh cycle

STATIC / DERIVED
8
Standards + scenarios

Data Source Registry — April 2026

OSM	OSM Power Infrastructure	WEEKLY	Node/edge	3 vars
BLS-LM	BLS Labour Market Statistics	MONTHLY	County	1 var
DOE	DOE/EIA (Department of Energy)	QUARTERLY	County	5 vars
EIA	EIA (Energy Information Administration)	ANNUAL	County	8 vars
Census	Census Bureau (US Census Bureau)	ANNUAL	County	8 vars
NERC	NERC (North American Electric Reliability Corp)	ANNUAL	Regional	2 vars
FEMA	FEMA / EPA (Federal Emergency Management Agency)	ANNUAL	County	4 vars
EEA	EEA Air Quality e-Reporting	ANNUAL	~1 km + station	3 vars
BLS	BLS (Bureau of Labor Statistics)	ANNUAL	County	3 vars
NTIA	NTIA Broadband Map	ANNUAL	County	2 vars
CISA	CISA (Cybersecurity and Infrastructure Security Agency)	ANNUAL	National	2 vars
FEMA-FR	FEMA Flood Risk / Hazard Mitigation	ANNUAL	County	2 vars
NRC	NRC (Nuclear Regulatory Commission)	ANNUAL	Site-level	1 var
BEA	BEA (Bureau of Economic Analysis)	ANNUAL	County	3 vars
EIA-SS	EIA Statistical Series	ANNUAL	State/County	2 vars
FRA	FRA (Federal Railroad Administration)	ANNUAL	County	1 var
DOT	DOT/FHWA (Dept of Transportation)	ANNUAL	County	1 var
FERC	FERC Monitoring Report	ANNUAL	Regional	2 vars
PUC	State PUCs (Public Utility Commissions)	ANNUAL	Service Territory	2 vars
CESER	DOE CESER (Cybersecurity, Energy Security, Emergency Response)	ANNUAL	Regional	2 vars
Eurostat	Eurostat Energy Statistics	ANNUAL	EU Regional	3 vars
JRC	JRC European Commission	ANNUAL	European	2 vars
USGS	USGS (US Geological Survey)	STATIC	County	3 vars
CDS	Copernicus CDS / ERA5	STATIC	0.25° (~25 km)	4 vars
DIM	Dimovski et al. (2025)	STATIC	County	3 vars
IEEE	IEEE C57.91 / IEC 60076	STATIC	Asset-level	16 vars

GFZ	GFZ German Research Centre	STATIC	~5 km	2 vars
USGS-SH	USGS Seismic Hazard Assessment	STATIC	County	1 var
Census-GEO	Census Bureau Geography Registry	STATIC	County	1 var
USGS-HZ	USGS Hazard Assessments	STATIC	~5 km	2 vars
ISO9223	ISO 9223 Corrosion	DERIVED	County	1 var
EIA-ISO	EIA/ISO-RTE (PJM/CAISO/MISO) — Data Portal	HOURLY	Balancing Authority	3 vars
OM	Open-Meteo Historical Weather	HOURLY	~1 km	3 vars
NOAA	NOAA/NWS (National Oceanic and Atmospheric Admin)	HOURLY	~1 km	2 vars
NOAA-NWS	NOAA/NWS (National Weather Service)	HOURLY	Station	3 vars
ENTSO-E	ENTSO-E Transparency	HOURLY	Cross-border	2 vars

C.1 Update Frequency Timeline

EXPECTED UPDATE WINDOWS

Weekly (2 sources) — OSM diffs + weather reanalysis

OSM Overpass · Open-Meteo/ERA5

Monthly (2 sources) — Grid load + procurement data

DOE Registry · EIA-ISO Transparency

Quarterly (1 sources) — DER registry + business data + macro indicators

DOE

Annual (19 sources) — Core grid quality + demographics + environment

EIA · Census · FEMA · BLS-LM · DOE · EIA-SS · FERC · EPA · Eurostat · BLS · NTIA · CISA · NOAA-NWS · USGS · NRC · FRA · BEA · PUC · CESER

Static (8 sources) — Standards + scenarios + seismic hazard

Dimovski · IEEE/IEC/CIGRÉ · Copernicus CMIP6 · GFZ · USGS · ISO 9223 · Census/OSM · USGS Seismic Hazard

C.2 Fleet Score Snapshot

FLEET SIZE
45,003
substations scored

MEDIAN R
0.395
P5=0.220 · P95=0.598

HIGH + CRITICAL
10510
23.4% of fleet

HIGH CONFIDENCE
100%

44981 subs · CI/R < 0.50

BAND DISTRIBUTION

Low 4720 (10.5%) | Medium 29773 (66.2%) | High 10507 (23.3%) | Critical 3 (0.0%)

No primary data sources have been updated since the v4.0.2 launch baseline. All **45,003 substation scores remain unchanged** this edition. The fleet median R of 0.395 (mean 0.400) reflects a distribution skewed toward the lower bands, with 10.5% in Low and 66.2% in Medium. The first material score changes are expected when FERC publishes the annual EIA Form 861 indicators (affecting C1–C4 and R6a) and when BEIS/DESNZ refreshes the Renewable Energy Planning Database (affecting T1). High-frequency sources (OSM, Open-Meteo) update weekly but feed only infrastructure topology (R4) and thermal proxy (I5), which have minimal impact on fleet-level score distribution.

C.3 Changelog — v3.4 → v4.0.2

18 changes tracked across P0, P1, and P2 releases

F1	NEW	New T component — Energy Transition Exposure (T1)	§2, §4
F2	NEW	New R6a — Restoration Speed Modifier (CAIDI-based)	§5
F3	ENHANCED	R3 enhanced — Energy Poverty Vulnerability (V_socio)	§5
F4	ENHANCED	R4 enhanced — Graph-theoretic betweenness + bridge detection	§5
F5	ENHANCED	R2 enhanced — Climate Trajectory (CMIP6 SSP2-4.5)	§5
F6	ENHANCED	R7 Digital Readiness — County-level NTIA Broadband model	§5
L1	NEW	New I5, I7–I9 metrics — thermal, load, corrosion, hydrogeological	§2, §8
L2	ENHANCED	E2 Innovation Enrichment — BEA + economic structure	§6
L3	ENHANCED	V_socio Fiscal Enrichment — Census + BLS + energy price	§5
L4	ENHANCED	R3 Demographic Shift Amplifier — Census net migration	§5
L5	DATA	95/95 variables operational (100%). 40 data sources total.	§8, §12
G1	DATA	DOE upgraded to quarterly registry — County-level DER registry	§12
G2	DATA	USGS upgraded to live API — County-level seismic data	§12
G3	DATA	OSM upgraded to Overpass API — 45,003 real substations (US)	§12
G4	NEW	R6b Seismic Hazard modifier — USGS Seismic Hazard classification	§5
G5	NEW	Network & Topology layer (H): 7 variables — EIA Grid Plan, seismic analysis	§12
G6	NEW	Output Scores layer (I): 7 variables — risk_score, ETTC, stationary probs	§12
G7	REGIONAL	US Data Sources — EIA, Census, USGS, FEMA, NRC, DOT, NERC, state PUCs	§12

SECTION D — MONTHLY DEEP-DIVE

The California Seismic Corridor: Where Seismic Risk Meets Energy Transition Pressure

Deep-dive rotation: Each edition features a substantive thematic analysis. The 12-month rotation: **Ed. 001** California seismic corridor · **002** DER saturation hotspots (East Anglia/West Midlands) · **003** Climate vulnerability under CMIP6 · **004** Metropolitan-Rural economic impact disparity · **005** Nuclear fleet dependency and phase-out risk · **006** Infrastructure age vs smart meter rollout · **007** Stress–Resilience quadrant trends · **008** T-component forward projections · **009** County Grid Health Cards · **010** Academic validation and peer feedback · **011** European replication feasibility · **012** Year in review.

D.1 Why the California, Why Now

CALIFORNIA SUBSTATIONS
774
202 HV · 572 MV

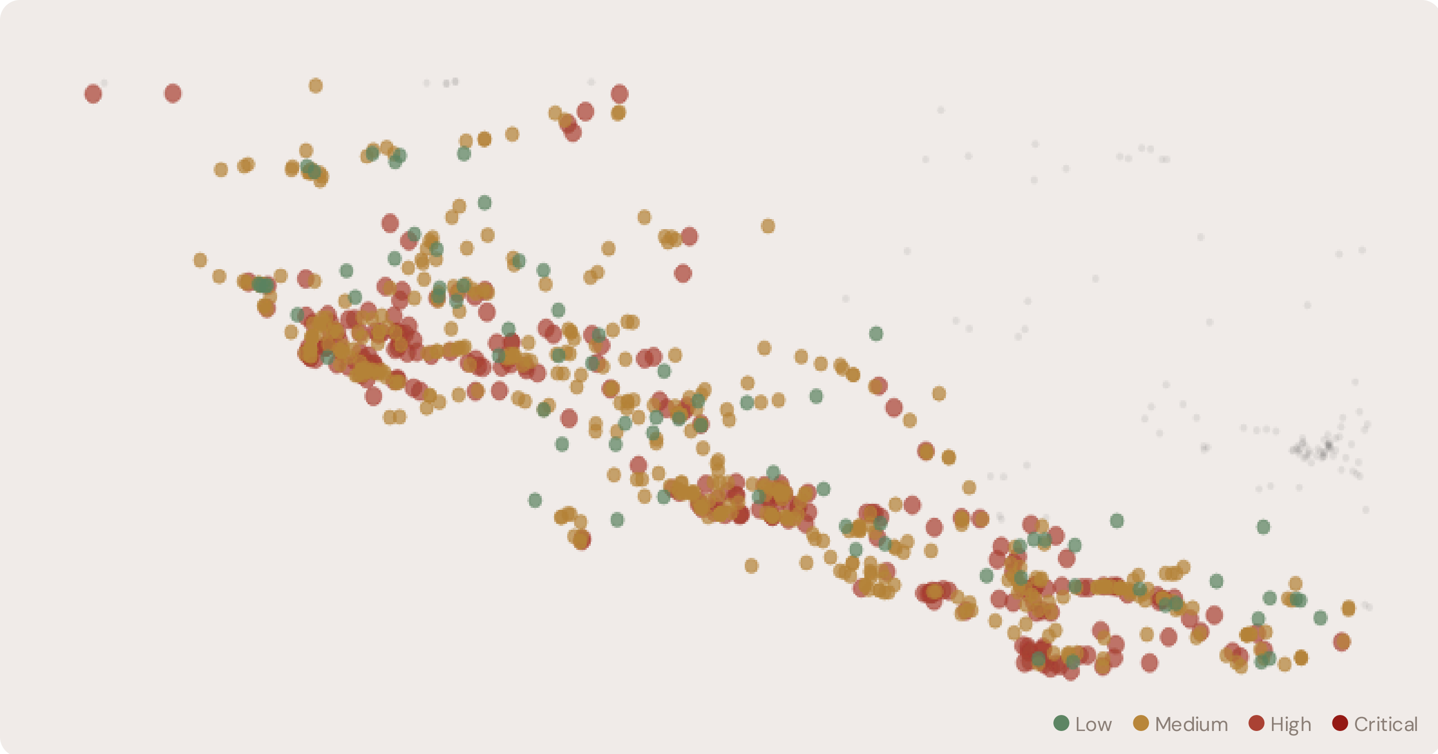
HIGH BAND
244
31.5% of corridor

CORRIDOR MEDIAN R
0.435
Fleet median: 0.395

WORST COUNTY
Kern
Worst sub: COPUS · R: 0.651

The California hosts **774 substations** across six counties, making it one of the most energy-critical corridors in the United States. However, its risk profile is disproportionately concentrated in the upper bands: **244 substations (31.5%)** are classified High, with only **75** in the Low band. 57% of California substations score above the national fleet median of 0.395. The corridor median R of 0.435 reflects the tension between seasonal demand volatility driven by tourism and agriculture, rapidly growing solar and wind capacity, and aging grid infrastructure not dimensioned for bidirectional renewable flows. The SSI flags continuity-of-supply deficits, infrastructure age mismatches with modern demand patterns, and grid modernization lag under federal 2030 timelines.

D.2 Corridor Map and Scoring



SUBSTATION	COUNTY	R_FINAL	BAND	C	V	I	E	S	T	ALERT
COPUS	California	0.651	High	0.893	0.413	0.690	0.373	0.575	0.623	C, I
AEC 300	California	0.646	High	0.867	0.232	0.521	0.264	0.874	0.569	C, S
TAP300074	California	0.645	High	0.894	0.247	0.557	0.406	0.573	0.704	C, T
UNKNOWN311117	California	0.644	High	0.899	0.272	0.479	0.349	0.791	0.533	C, S
UNKNOWN309973	California	0.644	High	0.880	0.114	0.514	0.068	0.910	0.646	C, S
KASSON-BANTA CARBONA JUNCTION	California	0.643	High	0.803	0.278	0.531	0.317	0.786	0.615	C, S
SWIFT	California	0.641	High	0.802	0.357	0.663	0.235	0.898	0.725	C, I, S, T
MOJAVE SIPHON	California	0.641	High	0.858	0.105	0.665	0.111	0.863	0.588	C, I, S
SHANDIN	California	0.640	High	0.851	0.123	0.600	0.151	0.869	0.542	C, S
CHICARITA	California	0.640	High	0.812	0.177	0.692	0.085	0.874	0.686	C, I, S, T

… 764 additional substations — explore full corridor in Digital Twin →

D.3 Component Analysis

COMPONENT AVERAGES — CALIFORNIA VS FLEET

C — Continuity 0.529 vs 0.505 (+5%)

MODIFIER AVERAGES — CALIFORNIA VS FLEET

R3 Consequence **0.955** fleet 0.955 ↓

I — Infrastructure	0.473 vs 0.552 (-14%)	
S — Saturation	0.485 vs 0.354 (+37%)	
V — Voltage	0.200 vs 0.261 (-23%)	
E — Economic	0.245 vs 0.129 (+90%)	
T — Transition	0.472 vs 0.447 (+6%)	

R4 Graph Crit.	0.981	fleet 0.980 ↓
R6a Restoration	1.021	fleet 1.020 ↑
R6b Seismic	1.001	fleet 1.000 →
R7 Digital Read.	0.996	fleet 0.995 →

The dominant risk driver across the California corridor is **T (Transition)**, averaging 0.472 against a fleet average of 0.447 — occupying 944% of its weight ceiling (w = 0.05). The second contributor is **E (Economic)** at 0.245 vs fleet 0.129 (245% of ceiling). Among modifiers, the R3 consequence multiplier averages 0.955 (fleet: 0.955), reflecting the socio-economic fragility of the California corridor's agriculture-intensive, tourism-dependent economy with growing renewable capacity but limited grid interconnection. The R6b seismic modifier averages 1.001, capturing the region's moderate but non-negligible seismic exposure — a dimension invisible to every competing framework.

D.4 County Breakdown

COUNTY RISK PROFILE — SORTED BY MEAN R

Longer bar = lower R = better resilience.

Unknown 774 subs · avg R = 0.421 · H:0 M:0

Loading county breakdown...

D.5 Implications

Loading implications...

SECTION E — RECURRING MODULE

American Context Card

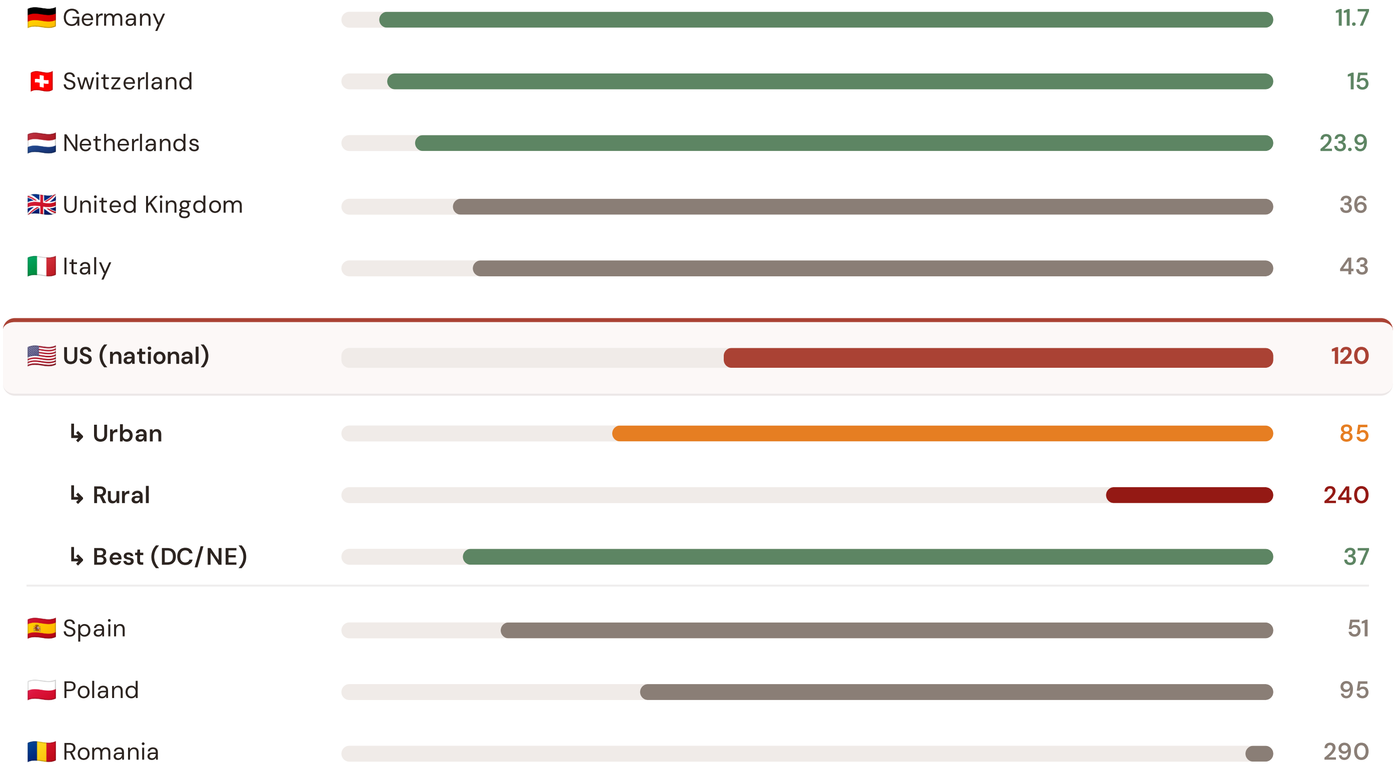
How this works: Each edition includes one American Context Card drawn from the §16 Peer Benchmarking framework. The card is selected to complement the month's deep-dive theme. This month: **SAIDI comparison** — because the California corridor analysis hinges on the US's continuity performance relative to European peers. Future editions rotate through: VoLL comparison (Ed. 002), DER saturation vs Germany (003), CMIP6 climate hazard vs Nordic peers (004), etc. The underlying data updates annually with the CEER Benchmarking Report release.

AMERICAN CONTEXT CARD — SAIDI BENCHMARKING

Ed. 001 · April 2026

Unplanned SAIDI (min/customer/year) — US vs International Peers

Longer bar = lower SAIDI = better reliability. Values in min/customer/year (excl. major events).



Source: EIA Form 861 (2023) for US national/urban/rural/state; CEER 7th Benchmarking Report (2022) for EU peers; Bundesnetzagentur (2024) for Germany. US values exclude major event days (IEEE 1366 standard). · See §16 Peer Benchmarking for full methodology. · Updated annually.

This month's key insight: The California corridor deep-dive shows **659 substations** (of 774) with C-component above 70% of its weight ceiling — indicating continuity-of-supply performance consistent with the US's rural territory classification.The California's average C of 0.529 is **1.0× the fleet average** of 0.505. In national terms, these substations individually perform closer to states with high SAIDI levels (200–400 min/customer/year) while the US national average stands at 120 min — well above EU leaders like Germany (12 min) and Switzerland (15 min). The SSI reveals what aggregate EIA statistics conceal: that the US's overall reliability position is not a uniform national condition but a statistical average of Northeastern-quality urban performance and rural/Western-level outage exposure — and the California seismic corridor concentrates the worst of the latter.

Regional Risk Ranking — All 52 States

Where does the California sit within the national landscape? Regions sorted by mean R_median, with band distribution and fleet comparison.

Longer bar = lower R = better resilience. Sorted worst → best.

#1 Washington — 372 subs · avg R = 0.522 · H:252

#2 New Jersey — 561 subs · avg R = 0.489 · H:239

#3 Maryland — 664 subs · avg R = 0.486 · H:296

#4 Massachusetts 483 subs · avg R = 0.474 · H:184

#5 Connecticut 255 subs · avg R = 0.467 · H:108

#6 Indiana 1340 subs · avg R = 0.465 · H:555

#7 Florida 1719 subs · avg R = 0.457 · H:732

#8 Oregon 226 subs · avg R = 0.451 · H:84

#9 Pennsylvania 790 subs · avg R = 0.448 · H:292

#10 Texas 1652 subs · avg R = 0.448 · H:658

#11 West Virginia 594 subs · avg R = 0.448 · H:216

#12 Illinois 1472 subs · avg R = 0.446 · H:518

#13 Georgia 1952 subs · avg R = 0.432 · H:684

#14 Delaware 154 subs · avg R = 0.427 · H:53

#15 District of Columbia 23 subs · avg R = 0.422 · H:6

#16 Hawaii 79 subs · avg R = 0.421 · H:18

#17 **California** 774 subs · avg R = 0.421 · H:244

#18 Rhode Island 99 subs · avg R = 0.416 · H:31

#19 New York 1594 subs · avg R = 0.413 · H:420 C:3

#20 Michigan 1243 subs · avg R = 0.413 · H:343

#21 Minnesota 2127 subs · avg R = 0.407 · H:554

#22 Utah 194 subs · avg R = 0.402 · H:24

#23 Ohio 1516 subs · avg R = 0.401 · H:379

#24 Idaho 13 subs · avg R = 0.400 · H:0

#25 Missouri 1541 subs · avg R = 0.395 · H:366

#26 Arizona 197 subs · avg R = 0.384 · H:43

#27 South Carolina 1597 subs · avg R = 0.382 · H:325

#28 Wisconsin 1408 subs · avg R = 0.381 · H:238

#29 Alaska 129 subs · avg R = 0.381 · H:15

#30 North Carolina 2720 subs · avg R = 0.380 · H:467

#31 Colorado 36 subs · avg R = 0.378 · H:6

#32 New Hampshire 80 subs · avg R = 0.377 · H:15

#33 Kentucky — 2192 subs · avg R = 0.377 · H:376

#34 Arkansas — 1028 subs · avg R = 0.377 · H:162

#35 Nevada — 101 subs · avg R = 0.377 · H:20

#36 Montana — 21 subs · avg R = 0.373 · H:1

#37 Alabama — 2549 subs · avg R = 0.371 · H:386

#38 Iowa — 1197 subs · avg R = 0.369 · H:149

#39 Louisiana — 1126 subs · avg R = 0.369 · H:141

#40 Virginia — 909 subs · avg R = 0.365 · H:123

#41 New Mexico — 273 subs · avg R = 0.365 · H:20

#42 Tennessee — 1522 subs · avg R = 0.360 · H:204

#43 Oklahoma — 2031 subs · avg R = 0.359 · H:171

#44 Kansas — 746 subs · avg R = 0.359 · H:82

#45 Wyoming — 16 subs · avg R = 0.359 · H:1

#46 Mississippi — 866 subs · avg R = 0.355 · H:99

#47 Maine — 172 subs · avg R = 0.353 · H:16

#48 Vermont — 82 subs · avg R = 0.352 · H:10

#49 North Dakota — 798 subs · avg R = 0.352 · H:54

#50 Nebraska — 895 subs · avg R = 0.349 · H:77

#51 South Dakota — 751 subs · avg R = 0.344 · H:43

#52 Unassigned — 124 subs · avg R = 0.321 · H:7

The California ranks **#17 of 52 states** by mean R_median, placing it among the upper tier of national risk. The three most stressed regions are Washington, New Jersey, Maryland, while the three most resilient are Unassigned, South Dakota, Nebraska. 24 of 52 states score above the fleet average of 0.400. This ranking — possible only because the SSI scores every substation individually rather than relying on aggregate DNO or national statistics — reveals the true spatial distribution of grid stress across UK. It is precisely this substation-level granularity that positions the SSI as a tool for Net Zero Strategy / Levelling Up Fund investment prioritisation: not treating entire regions as monoliths, but identifying the specific corridors within each that demand attention.

SECTION F — RECURRING MODULE

SSI Enhanced Neural Network — Monthly Topic

From grid intelligence to BESS valuation platform. The SSI Index answers *"How healthy is this substation?"* The SSI Enhanced Neural Network (SSI-ENN) extends this into: *"What is a BESS worth at this substation — to the NGESO/DNO and to the community?"* Each edition of this report explores one building block of the SSI-ENN's five-layer architecture, showing how the SSI Index data feeds directly into investment-grade grid analytics.

Layer 1 — Data	Layer 2 — Valuation	Layer 3 — Neural Net	Layer 4 — Coverage	Layer 5 — Platform
SSI v4.0 component taxonomy, normalisation, MC engine	NGESO/DNO + Community value stacks, TCO, real options	GBT+DNN ensemble, transfer learning, SHAP	Country prioritisation, DCI, registry, calibration	Products, API, revenue model, unit economics

LAYER 3 §0.5.6 · Inaugural edition · April 2026

Explainability & SHAP Values

Clients need to understand not just the prediction but WHY a substation ranks highly

Investment-grade models require interpretability. A BESS developer considering a £10M commitment needs to understand not just that a substation scores highly, but which specific factors drive that score — and whether those factors are likely to persist. SHAP (SHapley Additive exPlanations) provides a rigorous, game-theoretic attribution of each feature's contribution to each prediction. For every substation, the SSI-ENN produces a SHAP waterfall: starting from the fleet-average valuation and showing how each feature pushes the estimate up or down. This directly connects to the SSI Index: if SHAP identifies C (Continuity) as the dominant positive driver, the client knows the BESS value depends primarily on the substation's poor reliability history — and can verify this against FERC (Office of Gas and Electricity Markets)'s published EIA Form 861 data. Transparency is not optional; it is what converts model outputs into investment decisions.

KEY CONCEPTS

- → SHAP: game-theoretic feature attribution per prediction
- → Waterfall visualisation: fleet average → substation prediction
- → Direct traceability to SSI components and data sources
- → Feature importance validated against domain expertise

LIVE SSI DATA CONNECTION

The American training set comprises 45,003 substations with complete 95-feature vectors. Average CI_width of 0.147 across the fleet provides the uncertainty ground truth that the neural network's quantile regression heads must calibrate against. The fleet's risk distribution — 4720 Low, 29773 Medium, 10507 High, 3 Critical — ensures balanced training across all risk bands.

Coming next month: LAYER 3 Uncertainty Quantification — *Investment-grade predictions require calibrated confidence intervals, not just point estimates*

Looking Ahead

NEXT MONTH — EDITION 002

The Wales Solar Frontier

Deep-dive into Wales's grid risk profile — substation map, component analysis, county breakdown. American Context Card: Massive PV build-out vs weak transmission capacity and rural grid topology.

NEXT DATA REFRESH

Q3 2026 — 1 Quarterly Sources

DOE are due for refresh in Q3. Impact: BEIS/DESNZ (Renewable Energy Planning Database) data refresh schedule (T1), business fabric indicators (E2), macro-economic data. Highest-impact annual source pending: **EIA (Energy Information Administration)** (8 variables → C1–C4 (SAIDI/SAIFI/CAIDI), quality regulation).

FLEET SPOTLIGHT

0 Substations Approaching Critical Degradation

The SSI's 4-state Markov degradation model identifies **0 substations** with a stationary critical probability above 70% — meaning their long-term equilibrium state is dominated by degraded or critical condition. Without intervention, these substations will trend toward failure. Average estimated time-to-critical: ? years.

Edition 002 will be published on the second Thursday of May 2026. Deep-dive region: **Wales**. To ensure you receive it, confirm your subscription segment (General / Institutional / Academic) by email to ssi_index@ikenga.eu.

Feedback welcome. The SSI Index is designed to evolve through stakeholder dialogue. If you represent a American utility, FERC, NGESO, regional DNO, municipal energy company, or academic institution and would like to contribute data or methodology feedback, please contact us at ssi_index@ikenga.eu. The SSI's credibility depends on open scrutiny.

SSI Monthly Intelligence Report — Edition 001 (Inaugural)

Published 9 April 2026 · SSI Index v4.0.2 · 45,003 substations · 22,765 grid lines · 6 components · 20 metrics · 6 modifiers

10,000-iteration Gaussian Copula Monte Carlo · Sobol global sensitivity analysis (196,608 evals/substation) · CMIP6 SSP2-4.5